ARTIFICIAL INTELLIGENCE

Course Code: CS-202

Credit Hours: 3(2+1)

Pre-requisite: Nil

Course Objectives

The key objectives of this course include:

- To comprehend the fundamentals of Artificial Intelligence
- To evaluate and implement an appropriate uninformed/informed search algorithm for a problem and characterize its time and space complexity
- To translate natural language sentences (e.g., English) into logical statements
- To convert logic statements into a clause form and apply resolution to a set of logic statements to answer a query

Course Learning Outcomes

After successful completion of this course, the students should be able to:	GA	BT Level
1. **Discuss** the key concepts in the field of artificial intelligence	2	C2
2. **Implement** artificial intelligence techniques and case studies	3	C3
3. **Model** an automated solution for given problems using artificial intelligence paradigm.	4	C4

Course Contents

An Introduction to Artificial Intelligence and its Applications towards Knowledge-Based Systems; Introduction to Reasoning and Knowledge Representation, Problem Solving by Searching (Informed searching, Uninformed searching, Heuristics, Local searching, Minmax algorithm, Alpha-beta pruning, Game-playing); Case Studies: General Problem Solver, Eliza, Student, Macsyma; Learning from examples; ANN and Natural Language Processing; Recent trends in AI and applications of AI algorithms. Python programming language will be used to explore and illustrate various issues and techniques in Artificial Intelligence.

Mapping of CLOs to GA's

GA's/CLOs	CLO1	CLO2	CLO3
GA1 (Academic Education)			
GA2 (Knowledge for Solving Computing Problems)	✓		
GA3 (Problem Analysis)		✓	
GA4 (Design/Development of Solutions)			✓
GA5 (Modern Tool Usage)			
GA6 (Individual and Teamwork)			
GA7 (Communication)			
GA8 (Computing Professionalism and Society)			
GA9 (Ethics)			
GA10 (Life-long Learning)			

Prof. Dr. Aamir Hanif
Director Academics, Advanced Studies & Research
University Of Wah

Resources

Text Books:

1. Artificial Intelligence A Modern Approach by Stuart Russell and Peter Norvig, 4th edition, 2020

Reference Book:

Paradigms of Artificial Intelligence Programming: Case studies in Common Lisp", by Norvig, P., original edition, 1992.

AI algorithms, data structures, and idioms in Prolog, Lisp, and Java by Luger, G.F. and Stubblefield, W.A., 6th Edition, 2009.

Weekly Course Plan

Week No.	Topics
Week 1	**Introduction of Artificial Intelligence** • Basics of Artificial Intelligence (AI) • Course introduction • Scope, foundations and application Domain. • History of Artificial Intelligence • Importance of Turing Test and its Application
Week 2	**Intelligent Agents** • The Concept of Rationality • Properties of Task Environment • Structure of Agents
Week 3	**Problem solving Agents** • Understand about Search (Uninformed, Informed), its importance in AI and searching Algorithms.
Week 4	**Basic Algorithms for Problem Solving** • Breadth-first search • Uniform cost search • Depth-first search • Depth-limited search • Iterative deepening search • Best first search, beam search • A* Searching algorithms
Week 5	**Game Playing and Adversarial Search** • Cut off search • Pruning • Alpha beta Pruning **Case Studies:** • Game-playing (Chess, Checkers)
Week 6	**Expert Systems** • What is inference engine, and knowledge base? • Knowledge based agents • The Wumpus World Environment • Specifying the environment **Case Studies:** • General Problem Solver, Macsyma
Week 7	**Logic and Reasoning**

Prof. Dr. Aamir Hanif
Director Academics, Advanced Studies & Research
University Of Wah

	• Representation, Reasoning, and Logic • Representation • Inference **Case Studies:** • Logic-based Systems (Eliza, Student)
Week 8	**Mid Semester Examination**
Week 9	**Knowledge Representation** • Difference among Data, Information and Knowledge? • Why we need to differentiate among all terms? • Knowledge Representation (Logic-Based And Probabilistic-Based)
Week 10	**Constraint Satisfaction Problem** • CSP as a search problem • Local Search for Constraint Satisfaction Problem **Case Studies:** • Sudoku, Map Coloring
Week 11	**Prepositional Logic and First Order Predicate Logic** • Syntax • Semantics • Validity and inference • Models • Rules of inference for propositional logic
Week 12	**Introduction to Natural Language Processing (NLP)** • Supervised and Unsupervised Learning • Basics of NLP and its importance in AI • Text preprocessing techniques (tokenization, stemming, lemmatization) • Introduction to syntax and semantics in NLP
Week 13	**NLP Applications and Techniques** • Named Entity Recognition (NER) and its applications • Sentiment Analysis: Understanding opinions in text • Part-of-Speech (POS) Tagging: Identifying grammatical components
Week 14	**NLP and Machine Learning** • Integrating NLP with Machine Learning • Text classification and clustering techniques • Introduction to Word Embeddings and Word2Vec
Week 15	**Advance Topics NN and fuzzy systems** • Introduction to Neural Networks • Advanced topics in neural networks, like perceptron, and backpropagation Algorithms.

Prof. Dr. Aamir Hanif
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	• Introduction to Fuzzy and hybrid Intelligent systems
Week 16	**Advance Topics NN and fuzzy systems** • Introduction to Fuzzy and hybrid Intelligent systems • Recent trends in AI and applications of AI algorithms
End Semester Examination	

ARTIFICIAL INTELLIGENCE LAB

Course Code: CS-202L

Lab Objectives

The key objectives of this LAB include:

To acquaint students with a better understanding of Artificial Intelligence

To familiarize students with Python Programming

To equip students with hands-on experience in implementing Artificial Intelligence algorithms using Python Programming

Lab Learning Outcomes

After successful completion of this course, the students should be able to:	GA	BT Level
1. **Describe** the fundamental constructs of Python programming language.	2	P1
2. **Practice** artificial intelligence techniques and case studies.	5	P3
3. **Demonstrate** effectively as an individual or in a team to solve real-world problems.	6	A2 P4

Lab Contents

This lab course is about artificial intelligence (AI), which is one of the most advanced and complicated yet advantageous emerging technologies. In this lab, students will learn about AI algorithms from basic AI agents, graph search, and game playing using Python programming language. This lab's contents cover a wide range of fundamental AI topics, providing students with practical experience and exposure to various techniques and algorithms in the field of artificial intelligence.

Mapping of LLOs to GA's

GA's/LLOs	LLO1	LLO2	LLO3
GA1 (Academic Education)			
GA2 (Knowledge for Solving Computing Problems)	✓		
GA3 (Problem Analysis)			
GA4 (Design/Development of Solutions)			
GA5 (Modern Tool Usage)		✓	
GA6 (Individual and Teamwork)			✓
GA7 (Communication)			

Prof. Dr. Aamir Hanif
Director Academics, Advanced Studies & Research
University Of Wah